

The Virtual Bingo Blower: An open-source tool to generate ambiguity and risk in experiments.*

Ola Andersson[†] Geoffrey Castillo[‡] Erik Wengström[§]

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This paper introduces the Virtual Bingo Blower (VBB), an open-source tool to generate risk and ambiguity in computerized experiments. Using a physics engine—a computer simulation of a physical system—the VBB simulates a conventional bingo blower. Different aspects of the VBB, such as the number of balls, their colour, and their speed, can be easily modified. In an online experiment, we measure ambiguity attitudes and vary the source of ambiguity, using either the VBB or natural events. We find that the VBB and natural events result in a similar degree of ambiguity aversion. Further, we find that, by manipulating the number of balls, the VBB can be used to manipulate the perceived level of ambiguity.

Keywords: ambiguity, bingo blower, experimental tools, online experiment, randomization, risk

JEL Codes: C91, D81

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[†]Department of Economics, Uppsala University. ola.andersson@nek.uu.se

[‡]BEAR and Department of Economics, Nottingham Trent University. geoffrey.castillo@ntu.ac.uk

[§]Department of Economics, Lund University. erik.wengstrom@nek.lu.se

1 Introduction

Ambiguity refers to situations in which the probabilities attached to the outcomes are unknown. We often make decisions under ambiguity, for example when looking for a job, choosing a medical treatment, or investing in the stock market. Due to its importance, many experiments have been conducted to test theories of choice under ambiguity or to measure attitudes towards ambiguity (Camerer and Weber, 1992; Trautmann and van de Kuilen, 2015). A key question when designing such experiments is how to generate ambiguity.

We present a new method for generating ambiguity in experiments: the Virtual Bingo Blower (VBB). The VBB is inspired by physical bingo blowers, which have recently become popular for generating ambiguity in experiments (Georgalos, 2019, 2021; Hey et al., 2010; Hey and Pace, 2014; Kothiyal et al., 2014). In a bingo blower, balls of different colours move continuously, and a randomly selected ball determines the outcome. The constant motion of the balls makes it difficult to assess the proportions of colours, thereby generating ambiguity. Bingo blowers are thus an intuitive and transparent way of generating ambiguity. However, bingo blowers are bulky, noisy, and ill-suited for online experiments.

The VBB addresses the limitations of physical bingo blowers by being an open-source tool that runs directly in a web browser. This makes it particularly well-suited for online experiments as well as computerized laboratory experiments. The main advantage of the VBB over other non-physical devices is the use of a real-time physics engine that simulates the motion of balls according to physical laws. The physics simulation runs locally on participants' computers, independent of any input from experimenters. As a result, the movement of the balls is unpredictable and cannot be manipulated or anticipated by anyone, ensuring that ambiguity remains transparent and credible.

The other advantage of the VBB is that the perceived level of ambiguity can be easily controlled by adjusting features of the VBB such as the number of balls, their color, or even their speed of movement. With the VBB these changes are simple and can even be done in-between trials. Further, by revealing or using a number of balls that is easily countable, the VBB can transition from generating ambiguity to generating risk. The VBB is thus a flexible tool for generating both ambiguous and risky environments. It also allows researchers to avoid relying on

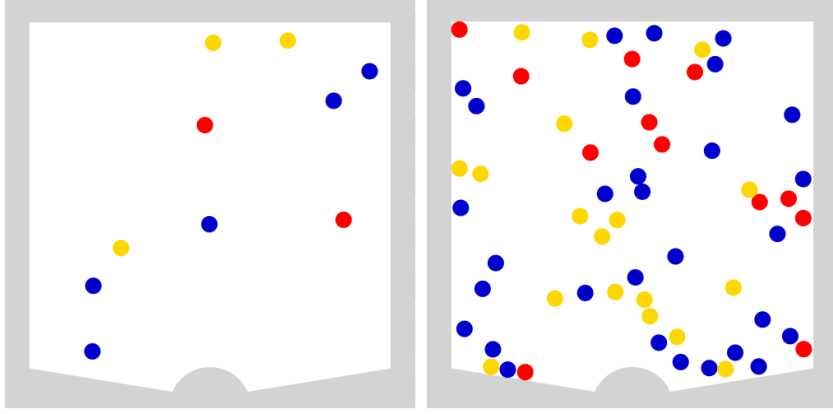


Figure 1: VBB with 10 balls and with 60 balls.

opaque random number generators, offering a more transparent alternative for randomization procedures. For instance, experimenters can use the VBB to play the lottery selected for payment at the end of a risk elicitation experiment, or to choose which round of a public goods game is relevant for payment.

We illustrate the usefulness of the VBB in an online experiment where ambiguity is generated either via the VBB or through natural events of the kind used by Baillon et al. (2018). Apart from testing the VBB, this design also allows us to study the impact of changing the source of ambiguity (Lotito et al., 2024; Watanabe and Fujimi, 2024). We find that both sources elicit similar degrees of ambiguity aversion, which suggests that subjects do not see the VBB as an artificial source of ambiguity. Moreover, we show that the perceived level of ambiguity can be manipulated by increasing the number of balls in the VBB.

Figure 1 shows screenshots of the VBB, for a small and a large number of balls. To see the VBB in action, a demo is available at <https://geoffreycastillo.com/bingo-blower-js-demo>. The VBB runs in all modern web browsers, including on mobile. It can thus be used in any web-based experimental software, such as Qualtrics, oTree (Chen et al., 2016), or LIONESS (Giamattei et al., 2020). Apart from being able to add balls of different colours, the parameters of the physics simulation, such as the air resistance or the bounciness of the balls, can also be adjusted. The VBB is open source and available as `bingo-blower.js` on GitHub at <https://github.com/geoffreycastillo/bingo-blower-js>, where we also provide documentation.

We present the design of the online experiment in Section 2. In this experiment, we elicit ambiguity attitudes using the method of Baillon et al. (2018). This method measures ambiguity attitudes via two indices: the b -index captures ambiguity aversion, while the a -index captures likelihood insensitivity and ambiguity perception. If there is no ambiguity at all, which would happen if subjects know the proportions of the different colours, then both indices should be equal to 0.

Across treatments, ambiguity is generated either using the VBB or natural events. In the VBB case, we further manipulate the number of balls in the VBB while keeping the proportions of the different colours constant. For the natural events, we use the evolution of the Dow Jones Industrial Average.

We present our results in Section 3. In line with our preregistered hypotheses, we find that the b -index is similar across treatments. We also find that it is slightly negative, which indicates that subjects are slightly ambiguity seeking. Previous studies have found markedly different ambiguity preferences between artificial and natural sources of ambiguity. Typically, artificial sources lead to clear ambiguity aversion while natural sources lead to ambiguity attitudes ranging from slightly ambiguity averse to slightly ambiguity seeking (Baillon et al., 2018; Lotito et al., 2024; Watanabe and Fujimi, 2024). The fact that we observe similar levels between the VBB and natural events suggests that the VBB was not perceived as an artificial source of ambiguity. The difference we do observe is that the b -index is more dispersed with the natural source of ambiguity.

Turning to the a -index, we find it is higher with natural events than with the VBB, indicating that subjects are more likelihood insensitive under the former source. We also find that increasing the number of balls in the VBB moderately increases the a -index and thus the perceived level of ambiguity. The increase is small, but our manipulation—increasing the number of balls—is deliberately simple. Although the effect is modest, its presence is noteworthy given the light manipulation and the earlier evidence that ambiguity levels are generally hard to shift within a given source (Garcia et al., 2020).

We conclude in Section 4 by addressing a potential issue that arises with the use of a virtual device such as the VBB to generate ambiguity: subjects’ suspicion and mistrust. Looking at the literature, we show that the available evidence indicates that physical and non-physical devices, or devices with different levels of manipulability, are perceived similarly by subjects. Further, we do not find evidence

of suspicion in our data when looking at the open-ended questions answered by subjects at the end of the experiment. Finally, we offer a number of ways to use the features of the VBB to mitigate any remaining suspicion.

Related literature. We contribute to the literature by offering a tool that shares the positive features of bingo blowers and adds the ease of computerized implementation. As Hey et al. (2010) explain, bingo blowers solve many of the issues associated with other methods used to generate ambiguity (see Camerer and Weber, 1992, and Trautmann and van de Kuilen, 2015, for reviews of the ambiguity literature). For example, they do not rely on information asymmetry between experimenters and subjects, as is the case with unknown urns (made famous by Ellsberg, 1961). Subjects can always look at the balls moving inside, and can witness how a ball is selected for payment. Bingo blowers also do not rely on second-order probability distributions to generate ambiguity (e.g., MacCrimmon and Larsson, 1979), which generate risk instead (Abdellaoui et al., 2011). Further, unlike natural events, bingo blowers are not affected by specific knowledge that subjects may have, which would need to be controlled for.

Due to the appeal of bingo blowers, other attempts have been made to translate them into computerized environments. For example, Levati and Morone (2013) used a 3D animation of a bingo blower cage projected on a wall to generate ambiguity in their experiments. Closer to us, Morone and Caferra (2024) also developed a new tool, the Ambiguity Box, which was used by Caferra et al. (2023) to study dynamic inconsistency under ambiguity. A crucial difference between the Ambiguity Box and the VBB is that the VBB uses a real-time physics engine to simulate the movement of the balls inside the bingo blower. By contrast, the Ambiguity Box is a sequence of pre-rendered static images displayed as a GIF, where the proportion of squares of different colours corresponds to the underlying probabilities. The VBB is thus less abstract, closer to the experience of a real bingo blower. Another implication of using a real physics simulation is that subjects can themselves trigger and witness the drawing of a ball from the bingo blower to determine their earnings. Neither subjects nor experimenters can predict which ball is going to be drawn. Experiments using the Ambiguity Box instead have to rely on a random number generator to play out the chosen lottery at the end of the experiment, which reduces transparency to subjects and could introduce mistrust, something the use of the

bingo blower initially aimed at avoiding.

2 Experimental design

The aim of the experiment is to illustrate the use of the VBB in a setting with ambiguity. To do so, we measure ambiguity attitudes in three treatments that differ in how we generate ambiguity. We then compare the measured ambiguity attitudes between these treatments.

2.1 Measuring ambiguity attitudes

We follow the method introduced by Baillon et al. (2018) to measure ambiguity attitudes. We refer to that paper for details.

The method starts by partitioning the set of possible events into at least three mutually exclusive and exhaustive, non-null, single events: $\{E_1, E_2, E_3\}$. The unions of these events are called composite events: $\{E_1 \cup E_2, E_1 \cup E_3, E_2 \cup E_3\}$. Then, for each event $E \in \{E_1, E_2, E_3, E_1 \cup E_2, E_1 \cup E_3, E_2 \cup E_3\}$, subjects are asked to report the matching probability m that makes them indifferent between receiving x if E is realised and receiving x with probability m . Throughout our experiment, we use $x = \text{£}3$.

Baillon et al. (2018) elicit matching probabilities using choice lists, which we also do. For each event, subjects face 20 choices presented as rows in a table. In each choice, subjects choose between an ambiguous option—winning $\text{£}3$ if the event is realized—and a risky option that offers a probability p of winning $\text{£}3$. Across all 20 choices, the ambiguous option remains the same while p varies between 0% and 100%. Since there are six events and 20 choices per event, this procedure creates 120 questions organized in six tables, one for each event. Figure 2 shows an example of such a table. As in Baillon et al. (2018), the tables are centered around 33% for single events and around 66% for composite events.

Based on these matching probabilities, Baillon et al. (2018) define two indices: the b -index and the a -index. The b -index measures aversion to ambiguity. As such, it captures motivational components of ambiguity attitudes. Ambiguity neutrality corresponds to $b = 0$, ambiguity aversion to $b > 0$, and ambiguity seeking to $b < 0$. The a -index measures likelihood insensitivity, which generalizes

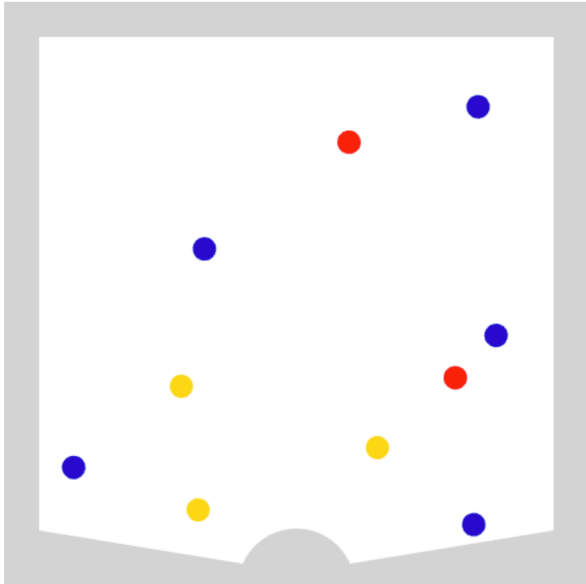
Option A		Option B	
You win £3 if one of the following balls is drawn:		You win £3 with the following chance:	
		A B 0 %	
		A B 20 %	
		A B 35 %	
		A B 40 %	
		A B 45 %	
		A B 50 %	
		A B 55 %	
		A B 60 %	
		A B 65 %	
		A B 70 %	
		A B 75 %	
		A B 80 %	
		A B 85 %	
		A B 90 %	
		A B 93 %	
		A B 95 %	
		A B 97 %	
		A B 98 %	
		A B 99 %	
		A B 100 %	

Figure 2: Example of a task (VBB10 treatment).

Table 1: Treatments.

Treatment	Source	Events
VBB10	VBB 10 balls	Ball drawn $\in \{\text{Blue, Red, Yellow}\}$
VBB60	VBB 60 balls	Ball drawn $\in \{\text{Blue, Red, Yellow}\}$
Natural	Dow	% change $\in \{(-100, -51), (-51, 0.11), (0.11, \infty)\}$

ambiguity perception. Higher values of a reflect a higher perceived level of ambiguity. Likelihood insensitivity corresponds to $a > 0$, while ambiguity neutrality gives $a = 0$. This index captures cognitive components of ambiguity attitudes and is thus more malleable.

2.2 Treatments

Between our treatments, we manipulate the source of the events. Table 1 summarizes our treatments.

2.2.1 The VBB treatments: VBB10 and VBB60

In the VBB treatments, an event is the colour of the ball drawn from the VBB. We use three different colours: red, blue, and yellow. The six events are thus: $\{\text{red, blue, yellow, red} \cup \text{blue, red} \cup \text{yellow, blue} \cup \text{yellow}\}$.

The proportion of each colour is the same in the VBB treatments: 20% red, 50% blue, and 30% yellow. The VBB treatments differ in the total number of balls in the VBB: 10 balls for VBB10, and 60 for VBB60. In VBB10, it may be possible to count balls in the VBB and thus know the proportions. By contrast, in VBB60, it is not possible to count the balls, and the proportions can only be guessed. Therefore, VBB10 generates risk while VBB60 generates ambiguity.

2.2.2 The Natural treatment

In the Natural treatment, an event is the daily percentage change of the Dow Jones Industrial Average Index on a specific day in the future. We use historical data on the daily movements of the Dow to calibrate the single events to be approximately equal in terms of probability to the single events in the VBB treatments. The partition that results from this calibration is the intervals of

percentage changes: $(-100, -0.51]$, $(-0.51, 0.11]$, and $(0.11, \infty)$. The six events are thus: $\{(-100, -0.51], (-0.51, 0.11], (0.11, \infty), (-100, -0.51] \cup (-0.51, 0.11], (-100, -0.51] \cup (0.11, \infty), (-0.51, 0.11] \cup (0.11, \infty)\}$.

2.3 Hypotheses

Our hypotheses relate to how the treatments affect the b - and the a -indices.

If the b -index strictly captures motivational components of ambiguity attitudes, it should not be affected by changes in the source of ambiguity. However, it is plausible that ambiguity aversion, and consequently the b -index, could vary with the treatment. We thus adopt a conservative stance and assume the b -index will not be affected by our treatments.

For the a -index, we expect that, moving from VBB10 to VBB60 and then to Natural, the level of perceived ambiguity to increase: while it may be possible to count the number of balls in VBB10, it is impossible in VBB60, and there is no probability distribution in Natural.

This reasoning can be summarized in the following two preregistered hypotheses, where we let $a(i)$ and $b(i)$ denote the a - and the b -index in treatment i .

Hypothesis 1 (b -index). Ambiguity aversion, as measured by the b -index, does not vary between treatments: $b(\text{VBB10}) = b(\text{VBB60}) = b(\text{Natural})$.

Hypothesis 2 (a -index). The perceived level of ambiguity, as measured by the a -index, increases from the VBB10 treatment to the VBB60 treatment, and reaches its highest level in the Natural treatment: $a(\text{VBB10}) < a(\text{VBB60}) < a(\text{Natural})$.

2.4 Recruitment and experimental procedures

Power calculations showed that we needed 150 subjects in each of the three treatments (significance level $\alpha = 0.05$, power $1 - \beta = 0.8$, effect sizes as in Baillon et al., 2018). Therefore, we aimed to recruit 450 subjects in total.

We recruited participants on Prolific. They were paid a fixed fee of £1 for participating in the experiment. In addition, participants could earn a bonus of £3 if a winning event occurred. Participants were required to live in the UK, speak English fluently, and have a Prolific approval rating of at least 95%.

The experiment was programmed using oTree (Chen et al., 2016).¹ The full instructions can be found in Appendix A. There we also present screenshots of all tasks. Our design, hypotheses, and exclusion criteria were preregistered on the OSF.²

481 subjects finished the experiment. We excluded one subject who did not do the experiment on a laptop or desktop as instructed, as well as 14 subjects whose screen had a refresh rate lower than 50 FPS—both exclusion criteria were pre-registered. That leaves us with 466 subjects: 162 in VBB10, 155 in VBB60, and 149 in Natural.³ In the regression analyses—which include a gender dummy—we additionally exclude three participants who preferred not to state their gender.

3 Results

In Figure 3 we provide an overview of the data. At first glance, we can see that, on average, the b -index is slightly negative in all treatments, which indicates slight ambiguity seeking behavior. The a -index is positive, which indicates that subjects perceived ambiguity and were likelihood insensitive.

To test our hypotheses, we follow Baillon et al. (2018) and use seemingly unrelated regressions (SUR). Table 2 reports the regression results. Model (1) does not include controls, and model (2) includes preregistered controls (age and gender). Note that the significance levels reported in this Table refer to two-sided tests. Since Hypothesis 2 is one-sided, we will report the associated one-sided p -values in the text. As a summary of our results, Figure 4 shows the values of a - and b -indices inferred from model (2).

As we can see, we do not observe treatment effects for the b -index. In both models, $b(\text{VBB10})$ is not statistically different from $b(\text{VBB60})$ and from $b(\text{Natural})$. Therefore, in line with Hypothesis 1, we do not reject the null that the b -index is the same across treatments.

On the other hand, we observe an effect on the a -index. $a(\text{VBB60})$ is statistically greater than $a(\text{VBB10})$ (without controls: one-sided $p = 0.090$, with controls:

¹Our oTree code is available at <https://github.com/geoffreycastillo/otree-virtual-bingo-blower>.

²See <https://osf.io/dm3kq>.

³Appendix B reports the demographic characteristics of our sample.

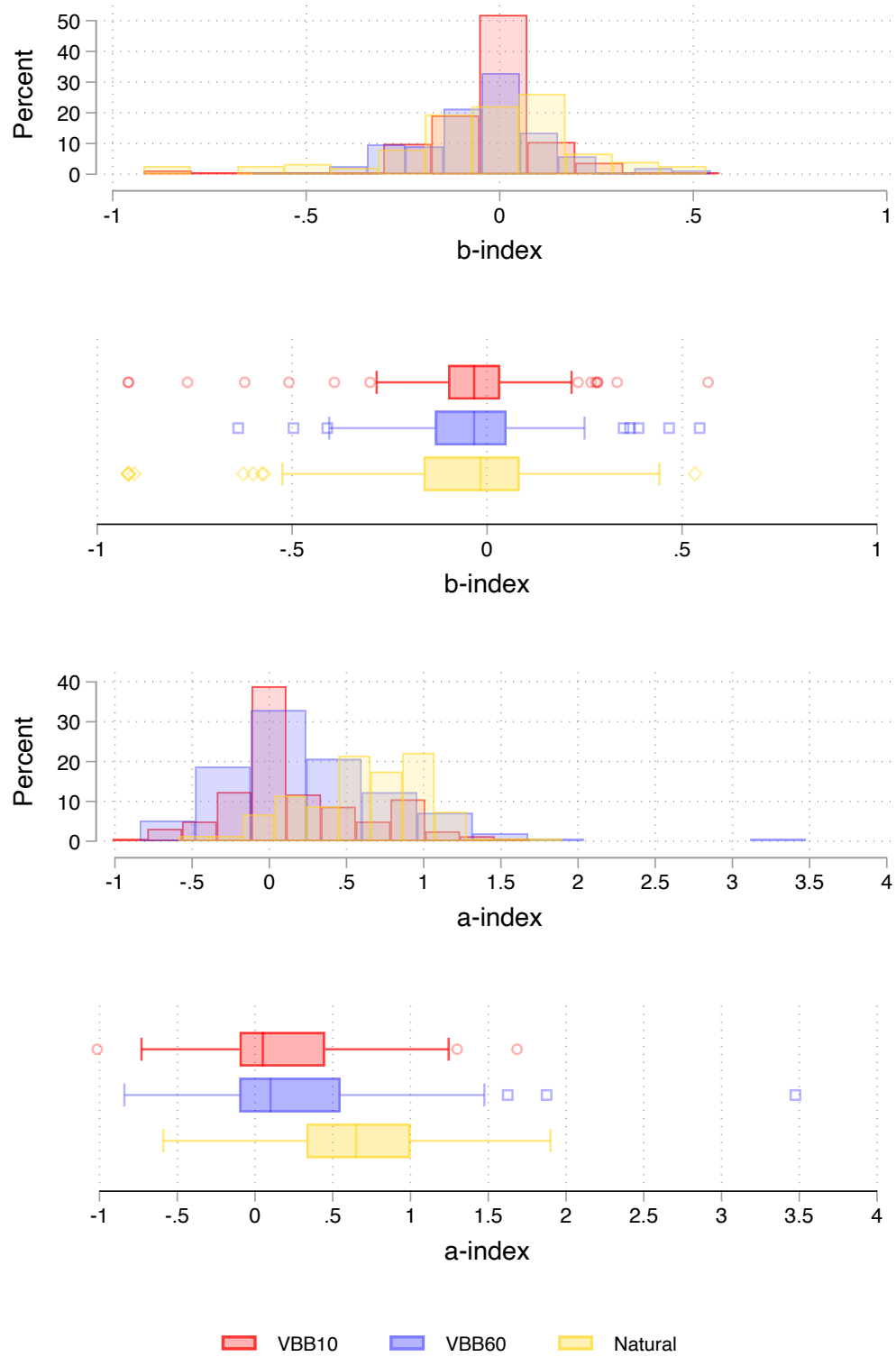


Figure 3: Histograms and box plots for the b - and a -index.

Table 2: Effect of treatments on a - and b -indices, SUR regression.

	(1)		(2)	
	a -index	b -index	a -index	b -index
VBB60	0.073 (0.055)	0.009 (0.023)	0.091 [•] (0.054)	0.008 (0.023)
Natural	0.449 ^{***} (0.055)	-0.011 (0.023)	0.454 ^{***} (0.055)	-0.009 (0.023)
Age category (ref.: 18-27)				
28-37			0.165 ^{**} (0.063)	0.008 (0.027)
38-47			0.037 (0.069)	-0.027 (0.029)
48-57			0.160 [*] (0.070)	0.011 (0.030)
58+			0.232 ^{**} (0.084)	-0.047 (0.036)
Female			0.037 (0.046)	0.014 (0.019)
Constant	0.176 ^{***} (0.038)	-0.047 ^{**} (0.016)	0.039 (0.064)	-0.050 [•] (0.027)
Observations	463		463	
R^2	0.138	0.002	0.163	0.012
Wald χ^2	74.133	0.733	90.311	5.502
Prob > χ^2	0.000	0.693	0.000	0.599
Breusch-Pagan χ^2	3.298		3.554	
Prob > Breusch-Pagan χ^2	0.069		0.059	

Notes. Standard errors in parentheses.

Two-sided p -values. [•] $p < 0.1$, ^{*} $p < 0.05$, ^{**} $p < 0.01$, ^{***} $p < 0.001$. One-sided p -values for the a -index are reported in the text.

Three participants who answered “prefer not to say” to the gender question are excluded from the regressions.

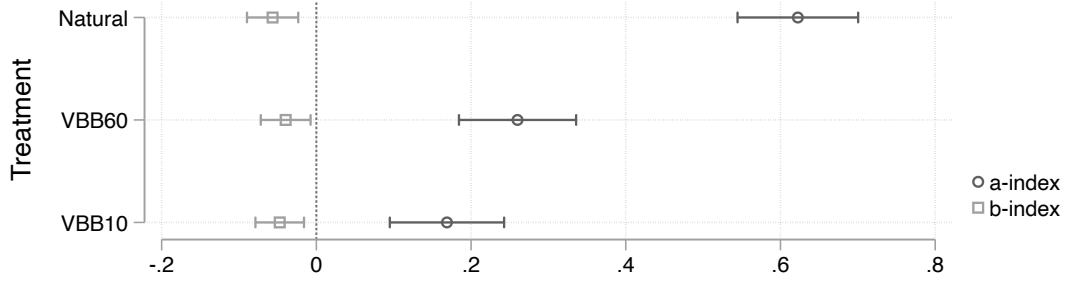


Figure 4: Predicted a -index and b -index estimates in each treatment with 95% confidence intervals (based on model 2, Table 2).

one-sided $p = 0.046$). $a(\text{Natural})$ is also statistically greater than both $a(\text{VBB10})$ and $a(\text{VBB60})$ (with and without controls: one-sided $p < 0.001$). Therefore, in line with Hypothesis 2, we reject the null that the a -index is the same across treatments: both the Natural treatment and the VBB60 treatment display higher a -index values compared to the VBB10 treatment.

The fact that only the a -index is affected by the treatments further confirms that these two indices measure different components of ambiguity attitudes. Baillon et al. (2018) already found that only the a -index is impacted by time pressure. We find a small negative correlation between the indices, significant at the 10% level (Pearson’s $\rho = -0.090$, $p = 0.051$). However, as can be seen in Figure 5, which displays the a -index and the b -index of each subject as a scatter plot, the indices are widely scattered and lack a clear pattern.

The values of the indices we observe in our experiment are consistent with those reported previously in the literature. For example, compared to Baillon et al. (2018), we also observe a slightly negative b -index, indicating ambiguity seeking behavior, and a positive a -index, indicating likelihood insensitivity. von Gaudecker et al. (2022) also observe likelihood insensitivity but, using a representative online sample—the LISS panel administered by CentERdata—they observe a slightly positive b -index, indicating ambiguity aversion. Both Baillon et al. (2018) and von Gaudecker et al. (2022) also found considerable heterogeneity in ambiguity attitudes, as we already saw in Figure 5.

In Appendix C we perform a number of additional analyses. First, we report pre-registered OLS regressions corresponding to the SUR reported in Table 2. We used SUR instead of OLS because the Breusch-Pagan test (also reported at

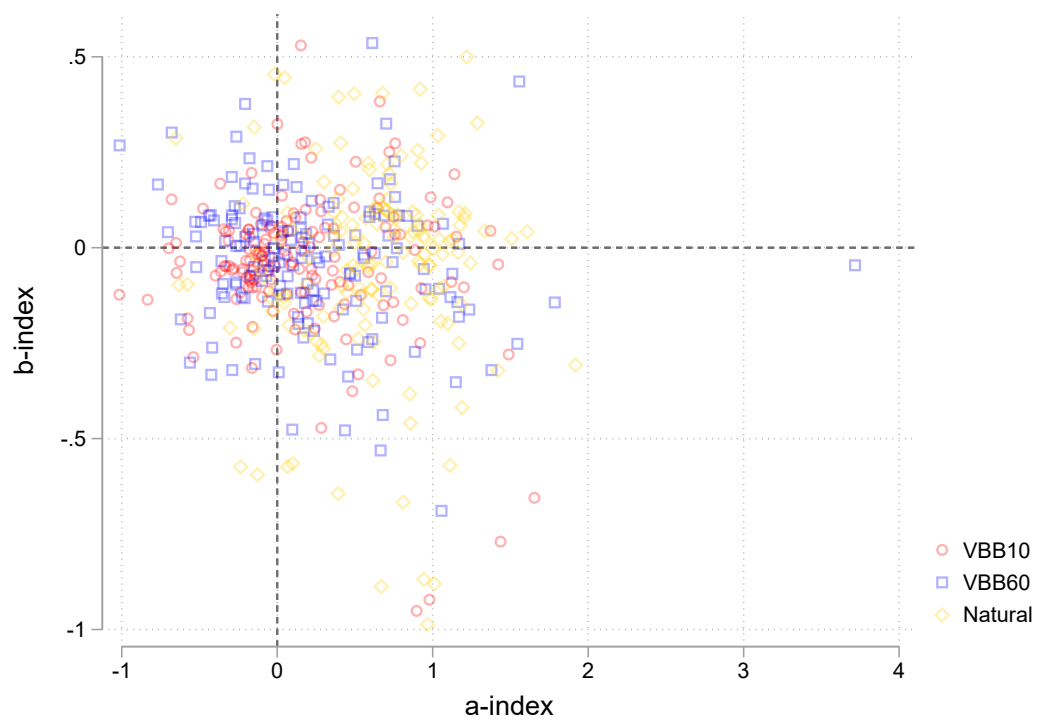


Figure 5: a -index and b -index estimates by treatment.

the bottom of Table 2) shows that the residuals from these OLS regressions are correlated, in which case SUR are preferred. Baillon et al. (2018) made the same observation and thus also used SUR. The main difference between SUR and OLS is that, under OLS, we reject the null that the impact of VBB10 and of VBB60 on the a -index is the same only at the 10% level and only with controls.

Second, we also report results from Mann-Whitney U tests, which we also preregistered. The test results corroborate the results from the SUR, except in the case of the a -index where we do not find a difference between VBB10 and VBB60 ($p = 0.454$). One reason for this difference may be that, as we could already see in Figure 3, the a -index is more dispersed in VBB60 than in VBB10. Indeed, unequal variances change the null of the Mann-Whitney U test from a median comparison to a comparison of the full distributions (Divine et al., 2018; Zimmerman, 2003). In the Appendix we confirm this intuition by following a procedure from Baillon et al. (2018) and studying how dispersion around the mean varies between treatments.

4 Discussion and conclusion

The VBB provides a transparent and flexible method for generating ambiguity and risk in computerized experiments. As we have shown, the VBB generates a similar degree of ambiguity aversion as natural events, suggesting that the VBB is not perceived as an artificial source of ambiguity. We have shown that increasing the number of balls in the VBB raises the perceived level of ambiguity, confirming that the tool can be used to introduce ambiguity in a controlled manner. While this manipulation was deliberately simple, more complex manipulations, such as increasing the speed of the balls, making their movement more erratic, or choosing colours that are less distinguishable, may be able to generate even greater differences.

A potential concern with the use of a virtual device such as the VBB to generate ambiguity is that it can suffer from subjects' mistrust. As the argument goes, any method to generate ambiguity that can be manipulated will be seen with suspicion by subjects who believe that the experimenter wants to deceive them, for example to save research money. We believe we can minimize or even address this concern in a number of ways.

First, contrary to the common mistrust argument, the evidence actually indicates that the potential for manipulability does not influence the elicited ambiguity attitudes. For example, Oechssler and Roomets (2015) show that using an Ellsberg urn that can be manipulated or a physical device that cannot—a Galton box—lead to the same ambiguity attitudes. Li et al. (2024) reach the same conclusion when comparing Ellsberg urns that have been constructed with different levels of manipulability.

Further, the experimenter could not manipulate the VBB even if they wanted to. As we have highlighted already, the VBB is powered by a real-time physics simulation that also runs on the subjects’ computers. As a consequence, the experimenter has no way to predict the movement of the balls. Also, the VBB has no function that can force a particular ball to be selected for payment, and even no function that could influence the draw, such as slowing down or speeding up the balls. Since the code is open source on GitHub, anyone can open it to verify this. We stressed all these points in our experiments.

To see if subjects believed us, we can look at the comments they left at the end of the experiment. In Appendix C.4 we classify such comments into different categories. Most of the comments are classified as engagement, understanding, or technical issues. We do not observe stark differences across treatments; if anything, subjects find the VBB to be more engaging than the Dow Jones. Importantly, we do not classify a single comment to a category that could remotely be associated with suspicion on the part of the subjects. We make the same observation across all the experiments in which we have used the VBB, totaling more than 1,000 subjects. Therefore, if any suspicion occurred, it must be very limited.

Finally, we can offer a number of practical solutions to mitigate any remaining fear of suspicion. As we did for our experiments, subjects can be convinced that they face a real physics simulation by allowing them to move the balls in the VBB and see how the physics simulation reacts to their movement. This ability can be kept right until a ball needs to be drawn for payment. In addition, some elements of the VBB can be chosen by the subjects themselves before the purpose of the VBB is revealed, mirroring Oechssler and Roomets’s (2015) construction of the Galton box. For example, subjects can be allowed to choose some of the parameters of the simulation, such as the speed of the wind or the weight of the balls. Subjects can be asked to provide numbers that determine how long the balls continue moving in

the VBB after clicking the button that starts the draw. The VBB can be further modified to allow subjects to click within the VBB to choose the location of the target from which the ball is drawn. With some or all of these additional steps, it becomes even more transparent that the experimenter has no influence on the movement of the balls in the VBB and thus on the draw of a ball.

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Appendices

Appendix A Instructions

When entering the experiment, subjects first saw a consent form. Then, they saw the instruction page.

A.1 Overall instructions

A.1.1 VBB10 and VBB60 treatments

Welcome! In this experiment you will be asked to make various decisions. Depending on your decisions you may earn a sizeable amount of money that will be paid to you after the experiment. This will be composed of the £1 participation fee plus your 'earnings' from the outcome of your decisions.

These earnings will depend partly on the decisions you take during the experiment and partly on chance. So you will need to read these instructions carefully.

Outline of the experiment You will be asked 6 questions of the same type. In each question you will be presented with two options, Option A and Option B.

After you have answered all 6 questions, one of them will be selected at random, and the Option you chose in this question will be played for real.

Therefore, it is in your interest to answer each question as if that were the question to be played out.

The Bingo Blower You will notice a virtual Bingo Blower below. You can see there are balls of three different colours: red, blue and yellow. It is a real physics simulation and not a pre-recorded video, meaning that we have no control on how the balls move.

To convince yourself that this Bingo Blower is simulated in real-time, on this page only you can manipulate the balls by clicking on one, dragging it and seeing how the other balls react. Later on you will not be able to control the balls to prevent you from affecting the outcome.

At the end of the experiment, in order to determine a colour, the Bingo Blower will be stopped and the ball closest to the center selected. Since the Bingo Blower is a real physics simulation, we cannot control the colour of the ball drawn.

You can use the buttons below to see how drawing a ball will work. After you click on 'Draw a ball', the balls keep tumbling for 3 seconds. Thereafter the balls are frozen and the ball closest to the center is chosen. If you want to try drawing a ball multiple times, click on 'Reset the Blower'.

[Figure A1]

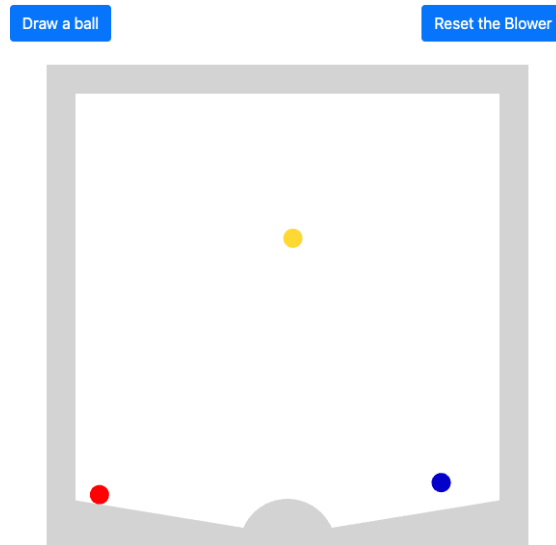


Figure A1

The ball drawn is: *[colour of the ball drawn]*

The questions In each question you will choose between two options, Option A and Option B:

- Option A (Bingo Blower option): you win £3 if the ball drawn from the Bingo Blower is of a certain colour, and nothing otherwise.
- Option B (given chance to win option): you have some chance to win £3.

You will be asked to state which one of these two options you prefer for various chances to win in Option B, from 0% to 100%. For example:

- If the chance to win in Option B is 100%, you will most likely prefer Option B because you are then guaranteed to win £3.
- If the chance to win in Option B is 0%, you will most likely prefer Option A because you might win something, as opposed to Option B where there is no chance to win at all.

You can think of the chance to win as the range of winning numbers in a bag containing numbers between 1 and 100. If the chance to win is 50%, it means you would win if a number between 1 and 50 is drawn from the bag, and you would lose if a number between 51 and 100 is drawn.

Example question The questions will look like this:

[Figure A2]

In this example, Option A would lead to a gain of £3 if the ball drawn from the Bingo Blower is red or yellow. Option B is presented with ascending chances to win. For each line we ask you to choose between Option A and Option B.

Option A You win £3 if one of the following balls is drawn:	Your choice	Option B You win £3 with the following chance:
 	☐ A ☐ B	0 %
	☐ A ☐ B	1 %
	☐ A ☐ B	2 %
	☐ A ☐ B	5 %
	☐ A ☐ B	10 %
	☐ A ☐ B	15 %
	☐ A ☐ B	20 %
	☐ A ☐ B	25 %
	☐ A ☐ B	30 %
	☐ A ☐ B	35 %
	☐ A ☐ B	40 %
	☐ A ☐ B	45 %
	☐ A ☐ B	50 %
	☐ A ☐ B	55 %
	☐ A ☐ B	60 %
	☐ A ☐ B	65 %
	☐ A ☐ B	70 %
	☐ A ☐ B	75 %
	☐ A ☐ B	85 %
	☐ A ☐ B	100 %

Figure A2

If you prefer Option A when, for instance, Option B offers a 40% chance to win, then you should also prefer Option A when Option B offers an even lower chance to win, say 25% . Similarly, if you prefer Option B with a 60% chance to win, then you should also prefer Option B with an even better chance to win, say 70%. Accordingly, in the figure above Option A is selected for every chance to win between 0 to 45%, and Option B, for every chance to win between 50% and 100%.

To save you the trouble of clicking for each line, you can click on Option A or Option B for one line and the computer will automatically fill your choices for the other lines. For example, in the figure above, button ‘B’ has been clicked only for the line corresponding to 50%.

You can change your choices as many times as you want. Once you are happy with your choices, click the ‘Next’ button at the bottom. You will not be allowed to go back to a question once you have clicked on the ‘Next’ button.

Payment After you have answered all 6 questions, one of them will be randomly selected by the computer. This question will be the question that counts for your payment. Then the computer will also select one of the lines:

- If, on this line, you had selected Option A, you will win £3 if the colour of the ball drawn from the Bingo Blower matches one of the colours displayed in Option A.
- If, instead, you had selected Option B, you will play that option. To do so the computer will randomly draw a number between 0 and 100. For example if Option B is ‘You win £3 with a 74% chance’ then you will win £3 if the

Option A You win £3 if the Dow evolves in one of the following ways:	Your choice	Option B You win £3 with the following chance:
- The Dow decreases by more than 0.51% - The Dow increases by more than 0.11%	☐ A ☐ B	0 %
	☐ A ☐ B	1 %
	☐ A ☐ B	2 %
	☐ A ☐ B	5 %
	☐ A ☐ B	10 %
	☐ A ☐ B	15 %
	☐ A ☐ B	20 %
	☐ A ☐ B	25 %
	☐ A ☐ B	30 %
	☐ A ☐ B	35 %
	☐ A ☐ B	40 %
	☐ A ☐ B	45 %
	☐ A ☐ B	50 %
	☐ A ☐ B	55 %
	☐ A ☐ B	60 %
	☐ A ☐ B	65 %
	☐ A ☐ B	70 %
	☐ A ☐ B	75 %
	☐ A ☐ B	85 %
	☐ A ☐ B	100 %

Figure A3

number drawn by the computer is strictly smaller than 74, and you will win nothing if it is equal to or larger than 75.

A.1.2 Natural treatment

[Identical to the instructions for VBB10 and VBB60, except:]

The questions In each question you will choose between two options, Option A and Option B:

- Option A (Dow option): you win £3 if the Dow (Dow Jones Industrial Average Index) increases or decreases by some percentages between the opening and closing price on *[date about a week after the experiment]*, and nothing otherwise.
- Option B (given chance to win option): you have some chance to win £3.

[Remaining of section ‘The questions’ identical]

Example question The questions will look like this:

[Figure A3]

In this example, Option A would lead to a gain of £3 if the Dow decreases by more than 0.51% or increases by more than 0.11%. Option B is presented with ascending chances to win. For each line we ask you to choose between Option A and Option B.

[Remaining of section ‘Example question’ identical]

Option A You win £3 if one of the following balls is drawn:		Your choice	Option B You win £3 with the following chance:
		A B	0 %
		A B	1 %
		A B	2 %
		A B	5 %
		A B	10 %
		A B	15 %
		A B	20 %
		A B	25 %
		A B	30 %
		A B	35 %
		A B	40 %
		A B	45 %
		A B	50 %
		A B	55 %
		A B	60 %
		A B	65 %
		A B	70 %
		A B	75 %
		A B	85 %
		A B	100 %

Figure A4

Payment After you have answered all 6 questions, one of them will be randomly selected by the computer. This question will be the question that counts for your payment. Then the computer will also select one of the lines:

- If, on this line, you had selected Option A, you will win £3 if the evolution of the Dow matches one of the descriptions displayed in Option A. We will use historical data from Yahoo Finance (available by clicking on this link: <https://finance.yahoo.com/quote/%5EDJI/history/>): the opening price is in the column 'Open', and the closing price in the column 'Close*'.
- If, instead, you had selected Option B, you will play that option. To do so the computer will randomly draw a number between 0 and 100. For example if Option B is 'You win £3 with a 74% chance' then you will win £3 if the number drawn by the computer is strictly smaller than 74, and you will win nothing if it is equal to or larger than 75.

A.2 Control questions

A.2.1 VBB10 and VBB60 treatments

To test your understanding we have designed the following control questions.

Imagine one of the questions is as follows:

[Figure A4]

Imagine further that this question is selected for payment at the end of the experiment.

Imagine that the line corresponding to a chance to win of 10% is selected. If your

choices are as described in the example above, which option would you play at the end of the experiment? [*Single choice*]

- Option A
- Option B

What happens if a red ball is drawn? [*Single choice*]

- I gain £0
- I gain £3
- It is irrelevant: how much I gain depends on the random number selected by the computer to play Option B.

Imagine that the line corresponding to a chance to win of 60% is selected. If your choices are as described in the example above, which option would you play at the end of the experiment? [*Single choice*]

- Option A
- Option B

What happens if a red ball is drawn? [*Single choice*]

- I gain £0
- I gain £3
- It is irrelevant: how much I gain depends on the random number selected by the computer to play Option B.

A.2.2 Natural treatments

To test your understanding we have designed the following control questions.

Imagine one of the questions is as follows:

[Figure A5]

Imagine further that this question is selected for payment at the end of the experiment.

Imagine that the line corresponding to a chance to win of 10% is selected. If your choices are as described in the example above, which option would you play at the end of the experiment? [*Single choice*]

- Option A
- Option B

What happens if the Dow decreases by more than 0.51%? [*Single choice*]

- I gain £0
- I gain £3
- It is irrelevant: how much I gain depends on the random number selected by the computer to play Option B.

Imagine that the line corresponding to a chance to win of 60% is selected. If your choices are as described in the example above, which option would you play at the end of the experiment? [*Single choice*]

- Option A
- Option B

Option A You win £3 if the Dow evolves in one of the following ways:	Your choice	Option B You win £3 with the following chance:
- The Dow decreases by more than 0.51% - The Dow increases by more than 0.11%	☐ A ☐ B	0 %
	☐ A ☐ B	1 %
	☐ A ☐ B	2 %
	☐ A ☐ B	5 %
	☐ A ☐ B	10 %
	☐ A ☐ B	15 %
	☐ A ☐ B	20 %
	☐ A ☐ B	25 %
	☐ A ☐ B	30 %
	☐ A ☐ B	35 %
	☐ A ☐ B	40 %
	☐ A ☐ B	45 %
	☐ A ☐ B	50 %
	☐ A ☐ B	55 %
	☐ A ☐ B	60 %
	☐ A ☐ B	65 %
	☐ A ☐ B	70 %
	☐ A ☐ B	75 %
	☐ A ☐ B	85 %
	☐ A ☐ B	100 %

Figure A5

What happens if the Dow decreases by more than 0.51%? *[Single choice]*

- I gain £0
- I gain £3
- It is irrelevant: how much I gain depends on the random number selected by the computer to play Option B.

A.3 Tasks

Subjects saw 6 questions, randomized independently for each subject.

A.3.1 VBB10 treatment

Figure A6 gives an example of a task for a simple event, and Figure A7, for a composite event

A.3.2 VBB60 treatment

Tasks in the VBB60 treatments were identical to those in the VBB10 treatment except that that the virtual bingo blower contained 60 balls.

A.3.3 Natural treatment

Figure A8 gives an example of a task for a simple event, and Figure A9, for a composite event

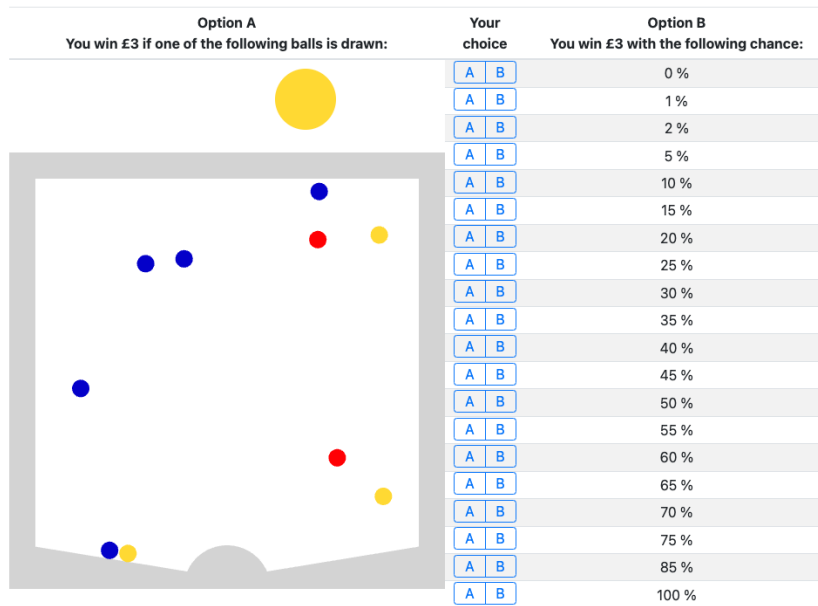


Figure A6: Example of a choice task in VBB10, simple event.

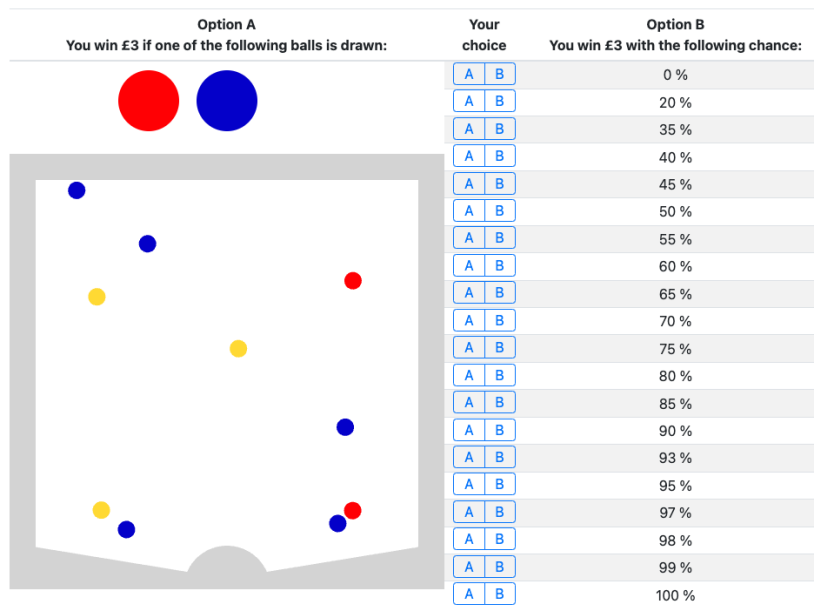


Figure A7: Example of a choice task in VBB10, composite event.

Option A You win £3 if the Dow evolves in one of the following ways:	Your choice	Option B You win £3 with the following chance:
The Dow increases by more than 0.11%	☐ A ☐ B	0 %
	☐ A ☐ B	1 %
	☐ A ☐ B	2 %
	☐ A ☐ B	5 %
	☐ A ☐ B	10 %
	☐ A ☐ B	15 %
	☐ A ☐ B	20 %
	☐ A ☐ B	25 %
	☐ A ☐ B	30 %
	☐ A ☐ B	35 %
	☐ A ☐ B	40 %
	☐ A ☐ B	45 %
	☐ A ☐ B	50 %
	☐ A ☐ B	55 %
	☐ A ☐ B	60 %
	☐ A ☐ B	65 %
	☐ A ☐ B	70 %
	☐ A ☐ B	75 %
	☐ A ☐ B	85 %
	☐ A ☐ B	100 %

Figure A8: Example of a choice task in Natural, simple event.

Option A You win £3 if the Dow evolves in one of the following ways:	Your choice	Option B You win £3 with the following chance:
The Dow decreases by less than 0.51% or increases by less than 0.11%	☐ A ☐ B	0 %
	☐ A ☐ B	1 %
	☐ A ☐ B	2 %
	☐ A ☐ B	5 %
	☐ A ☐ B	10 %
	☐ A ☐ B	15 %
	☐ A ☐ B	20 %
	☐ A ☐ B	25 %
	☐ A ☐ B	30 %
	☐ A ☐ B	35 %
	☐ A ☐ B	40 %
	☐ A ☐ B	45 %
	☐ A ☐ B	50 %
	☐ A ☐ B	55 %
	☐ A ☐ B	60 %
	☐ A ☐ B	65 %
	☐ A ☐ B	70 %
	☐ A ☐ B	75 %
	☐ A ☐ B	85 %
	☐ A ☐ B	100 %

Figure A9: Example of a choice task in Natural, composite event.

Appendix B Summary statistics on the sample

Table B1 describes the sample in terms of gender and age. As we can see there are subjects of varying ages present and slightly more females in the sample.

Appendix C Additional analyses

C.1 OLS regressions

We preregistered to run OLS regressions with and without controls. As mentioned in the main text, residuals from the estimation of the two indices reveal that these are correlated, thus warranting a SUR approach. For completeness, we report the results from the regressions here. Results are reported in Table C1 and corroborate the previous results.

The main difference between the SUR and the OLS regressions is that, under OLS and without controls, we cannot reject the null that the impact of VBB10 and of VBB60 on the a -index is the same (one-sided $p = 0.105$). However, in line with the SUR estimates, the differences become more pronounced with controls, and we reject the null at the 10% level (one-sided $p = 0.057$).

C.2 Mann-Whitney U tests

We also preregistered testing for treatment differences using Mann-Whitney U (MWU) tests. We report the results in Table C2 for the b -index and in Table C3 for the a -index.

The test results for the b -index fully corroborate our regression results, showing no difference between treatments. The MWU-test results of the a -index show that the distribution, in both the VBB10 treatment (p -value < 0.000) and the VBB60 treatment (p -value < 0.000), significantly differs from that of the Natural treatment, but not from each other. One reason for the difference in significance between the SUR and MWU test may be that the latter is ordinal and thus less sensitive to the dispersion of the index values. As noted in Figure 3, the VBB60 treatment clearly displays a higher degree of dispersion compared to VBB10. We study such dispersion in Appendix C.3

C.3 Analysis of dispersion of the indices

We further study the dispersion of the indices by following the procedure in Baillon et al. (2018). We take the absolute value of each index normalized around its treatment mean, and run a regression to see how the dispersion around the mean varies across treatments. To make it consistent with the previous regression analysis we use the same controls. In Table C4 we see a significant difference in dispersion

between the two VBB treatments for the a - but not b -index. Clearly, the VBB10 treatment causes less dispersion than the VBB60 treatment.

C.4 Comments categorization

The categorization of comments was initially carried out using a pattern-based classification approach, implemented with a machine learning model (ChatGPT) and Python.⁴ This approach goes beyond simple keyword spotting and takes into account contextual clues and patterns in the text. The categories were initially suggested by the language model and were subsequently manually reviewed and adjusted by the authors for accuracy. Based on this analysis, comments were assigned to one of the following categories:

- **Understanding:** Comments that mentioned confusion, difficulty in understanding, or explicitly stated incomprehension.
- **Engagement:** Comments that discussed interest, enjoyment, boredom, or difficulty in focusing on the experiment.
- **Technical Issues:** Comments that reported glitches, technical issues, or slow loading times.
- **Neutral/No comments:** Comments that explicitly stated no opinion or feedback.

We restrict attention to the 466 subjects used in the main analysis. A total of 87 comments were received in the experiment, out of which 19 were either neutral or had no comment. Table C5 provides a summary of the categorized comments received from participants across different treatment groups. Each category is followed by the number of comments corresponding to it, broken down by treatment type (VBB10, VBB60, Natural). To offer a glimpse into the nature of the comments, representative examples for each category are also included in the table.

⁴See <https://bit.ly/46gujYR> for the full set of ChatGPT prompts used for the classification.

Table B1: Proportion of the different demographic categories by treatment

	VBB10	VBB60	Natural
Age:			
18-27	0.22	0.26	0.21
28-37	0.29	0.30	0.28
38-47	0.16	0.21	0.22
48-57	0.21	0.18	0.16
58+	0.12	0.06	0.13
Sex:			
Male	0.41	0.45	0.38
Female	0.59	0.55	0.62
Prefer not to say	0.00	0.00	0.01
Observations	162	155	149

Table C1: Effect of treatments on a - and b -indices, OLS regressions.

	a -index	b -index	a -index	b -index
VBB60	0.073 (0.058)	0.009 (0.020)	0.091 (0.058)	0.008 (0.020)
Natural	0.449*** (0.050)	-0.011 (0.025)	0.454*** (0.050)	-0.009 (0.025)
Age category (ref.: 18-27)				
28-37			0.165* (0.065)	0.008 (0.026)
38-47			0.037 (0.062)	-0.027 (0.030)
48-57			0.160* (0.074)	0.011 (0.029)
58+			0.232** (0.085)	-0.047 (0.046)
Female			0.037 (0.045)	0.014 (0.019)
Constant	0.176*** (0.036)	-0.047*** (0.014)	0.039 (0.063)	-0.050• (0.030)
Observations	463	463	463	463
R^2	0.138	0.002	0.163	0.012

Notes. Robust standard errors in parentheses.

Two-sided p -values. • $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. One-sided p -values for the a -index are reported in the text.

Three participants who answered “prefer not to say” to the gender question are excluded from the regressions.

Table C2: b -index: means (on diagonal) and p -values from MWU-tests (off diagonal)

	VBB10	VBB60	Natural
VBB10	-0.047		
VBB60	0.774	-0.038	
Natural	0.438	0.657	-0.053

Table C3: a -index: means (on diagonal) and p-values from MWU-tests (off diagonal)

	VBB10	VBB60	Natural
VBB10	0.176		
VBB60	0.454	0.249	
Natural	0.000	0.000	0.622

Table C4: Effect of treatments on absolute value of a - and b -indices, SUR regression.

	(1)		(2)	
	$ a - mean $	$ b - mean $	$ a - mean $	$ b - mean $
VBB60	0.076 [*] (0.034)	0.019 (0.017)	0.082 [*] (0.034)	0.024 (0.016)
Natural	-0.019 (0.035)	0.074 ^{***} (0.017)	-0.016 (0.035)	0.072 ^{***} (0.017)
Age category (ref.: 18-27)				
28-37			0.004 (0.040)	-0.024 (0.019)
38-47			-0.027 (0.044)	-0.008 (0.021)
48-57			0.051 (0.044)	-0.017 (0.021)
58+			0.039 (0.053)	0.069 ^{**} (0.025)
Female			0.030 (0.029)	0.023 [•] (0.014)
Constant	0.357 ^{***} (0.024)	0.108 ^{***} (0.012)	0.327 ^{***} (0.041)	0.098 ^{***} (0.019)
Observations	463		463	
R^2	0.018	0.042	0.028	0.080
Wald χ^2	8.356	20.315	13.313	40.312
Prob > χ^2	0.015	0.000	0.065	0.000
Breusch-Pagan χ^2	7.931		7.284	
Prob > Breusch-Pagan χ^2	0.005		0.007	

Notes. Standard errors in parentheses.

• $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table C5: Categorization of comments.

Category	VBB10	VBB60	Natural	Total	Example Comments
Engagement	10	13	6	29	“ I found it interesting! really enjoy these types of surveys.”, 11I found this explanation very confusing”, “I took a shot but was unlucky ”
Neutral/No Comment	9	8	2	19	“ no", "No further comments", "None. ”
Technical Issues	2	3	3	8	“It was difficult to count the number of each coloured ball as they were jumping all the time. There was also slight delay for the program to respond to clicks”, “It was a bit slow to load each page so I was worried it would crash. Otherwise it was fine and pretty fun.”, “Very interesting study. The page seemed to load rather slowly at times (even on pages without the balls present) but everything worked with patience. ”
Understanding	10	8	13	31	“ I had absolutely no idea what was going on despite reading the instructions several times. The description of Option B was completely baffling to me”, “Although I did my best I apologize for struggling to fully understand the instructions. Thank you for giving me the opportunity to participate.”, “confusing ”
Total	31	32	24	87	